1. Code:

```c
#include <stdio.h>

struct Student {
    char name[50];
    int rollNumber;
    float marks[5];
    float total;
    float percentage;
    char grade;
};

void calculateTotalAndPercent(struct Student *s) {
    s->total = 0;
    for (int i = 0; i < 5; i++) {
        s->total += s->marks[i];
    }
    s->percentage = (s->total / 500.0) * 100.0;
}

char calculateGrade(float percentage) {
    if (percentage >= 90.0) {
        return 'A';
    } else if (percentage >= 75.0) {
        return 'B';
    } else if (percentage >= 50.0) {
        return 'C';
    } else if (percentage >= 35.0) {
        return 'D';
    } else {
        return 'F';
    }
}

void displayStudent(struct Student s) {
    printf("\n Student Result \n");
    printf("Name: %s\n", s.name);
    printf("Roll Number: %d\n", s.rollNumber);
    printf("Total Marks: %.2f / 500\n", s.total);
    printf("Percentage: %.2f%%\n", s.percentage);
    printf("Grade: %c\n", s.grade);
}

int main() {
    struct Student std;

    printf("Enter Student Name: ");
    scanf(" %[^\n]", std.name);

    printf("Enter Roll Number: ");
    scanf("%d", &std.rollNumber);

    printf("Enter marks for 5 subjects:\n");
    for (int i = 0; i < 5; i++) {
        printf("Subject %d: ", i + 1);
        scanf("%f", &std.marks[i]);
    }

    calculateTotalAndPercent(&std); std.grade = calculateGrade(std.percentage);
    displayStudent(std);

    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    ···        zsh

amol@Amols-MacBook-Pro PP-lab % gcc src/exp2/program.c
amol@Amols-MacBook-Pro PP-lab % ./a.out
Enter Student Name: Amol
Enter Roll Number: 63
Enter marks for 5 subjects:
Subject 1: 56
Subject 2: 54
Subject 3: 45
Subject 4: 47
Subject 5: 38

 Student Result
Name: Amol
Roll Number: 63
Total Marks: 240.00 / 500
Percentage: 48.00%
Grade: D
amol@Amols-MacBook-Pro PP-lab %
```

2. Code:

Program 2

```c
#include <stdio.h>

struct Account {
    int accountNumber;
    char name[50];
    float balance;
};

void deposit(struct Account *acc, float amount) {
    if (amount > 0) {
        acc->balance += amount;
        printf("Deposit successful. New balance: %.2f\n", acc->balance);
    } else {
        printf("Invalid deposit amount.\n");
    }
}

void withdraw(struct Account *acc, float amount) {
    if (amount > 0) {
        if (acc->balance >= amount) {
            acc->balance -= amount;
            printf("Withdrawal successful. New balance: %.2f\n", acc->balance);
        } else {
            printf("Insufficient balance.\n");
        }
    } else {
        printf("Invalid withdrawal amount.\n");
    }
}

void displayBalance(struct Account acc) {
    printf("\n Account Details \n");
    printf("Account Number: %d\n", acc.accountNumber);
    printf("Name: %s\n", acc.name);
    printf("Current Balance: %.2f\n", acc.balance);
}

int main() {
    struct Account userAcc;
    int choice;
    float amount;

    printf("Enter Account Number: ");
    scanf("%d", &userAcc.accountNumber);

    printf("Enter Account Holder Name: ");
    scanf(" %[^\n]", userAcc.name);

    printf("Enter Initial Balance: ");
    scanf("%f", &userAcc.balance);

    do {
        printf("\n1. Deposit\n2. Withdraw\n3. Balance Inquiry\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            printf("Enter amount to deposit: ");
            scanf("%f", &amount);
            deposit(&userAcc, amount);
        } else if (choice == 2) {
            printf("Enter amount to withdraw: ");
            scanf("%f", &amount);
            withdraw(&userAcc, amount);
        } else if (choice == 3) {
            displayBalance(userAcc);
        } else if (choice == 4) {
            printf("Exiting system...\n");
        } else {
            printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}
```

Output:

```
amol@Amols-MacBook-Pro PP-lab % g++ src/exp2/program2.c
clang++: warning: treating 'c' input as 'c++' when in C++ mode,
amol@Amols-MacBook-Pro PP-lab % ./a.out
Enter Account Number: 1
Enter Account Holder Name: Amol
Enter Initial Balance: 0

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit
Enter your choice: 1
Enter amount to deposit: 12
Deposit successful. New balance: 12.00

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit
Enter your choice: 3

 Account Details
Account Number: 1
Name: Amol
Current Balance: 12.00

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit
Enter your choice: 2
Enter amount to withdraw: 10
Withdrawal successful. New balance: 2.00

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit
Enter your choice: 3

 Account Details
Account Number: 1
Name: Amol
Current Balance: 2.00
```